$$\mathcal{L}_{I \to T} = -\frac{1}{N} \sum_{i=1}^{N} \ln \frac{e^{S_{ii}}}{\sum_{j=1}^{N} e^{S_{ij}}}$$

$$\mathcal{L}_{T \to I} = -\frac{1}{N} \sum_{i=1}^{N} \ln \frac{e^{S_{ii}}}{\sum_{j=1}^{N} e^{S_{ji}}}$$

$$\mathcal{L} = \tfrac{1}{2}(\mathcal{L}_{I \to T} + \mathcal{L}_{T \to I})$$